S1 Text. Supplementary methods

Supporting information for: Kaya-guided hybrid deep learning for leave-country-out estimation of national CO_2 emissions: a 112-country benchmark with country-clustered inference, conformal uncertainty, and explainable AI.

A. Notation

Table A. Symbols used in the article and in this text.

Symbol	Meaning
i, t, τ	Country; target year; a year inside the covariate window
$c_{i,t}$	Per-capita fossil CO_2 emissions (t CO_2 per person per year)
$y_{i,t}$	$\ln c_{i,t}$, the quantity that the models estimate
E, G, P	Primary energy per capita (kWh per person); GDP per capita (international dollars per person, 2011 prices); population (persons)
e, g, p	$\log_{10}$ of E, G and P
$s^{coal}, s^{oil}, s^{gas}$	Shares of fossil CO_2 emissions from coal, oil and gas (fractions)
s^{lc}	Low-carbon share of primary energy (fraction)
C, Y, Q	National totals in the Kaya identity: fossil CO_2 emissions, GDP and primary energy consumption
W, F	Window length (8 years); number of covariates (6, or 7 in Experiment C)
$x_{i,\tau}, X_{i,t}$	Covariate vector of one year; window of $W \times F$ values
μ_{tr}, σ_{tr}	Mean and standard deviation of a covariate over the training countries of a fold
$\tilde{x}, \tilde{X}$	Standardized covariate vector and window
β, β_0, λ	Ridge coefficients, intercept and penalty
$\beta_K, \beta_{K,0}, r_\theta$	Weights and intercept of the Kaya head; residual network with parameters θ
n, N	Index and number of held-out samples
$\hat{y}, \hat{c}$	Estimates of y and of $c = \exp(y)$
h_τ, c_τ	Hidden state and cell state of a recurrent layer at window step τ
α_τ	Attention weight of window step τ
$T, \acute{y}, \hat{\sigma}$	Number of Monte-Carlo dropout passes (60); predictive mean and standard deviation
$s, \hat{q}, n_{cal}$	Conformal score, calibration factor and number of calibration samples
ϕ_j	SHAP value of input j

B. Accuracy measures

With c_n the observed and $\hat{c}_n$ the estimated per-capita emissions of held-out sample n, N the number of held-out samples and $\acute{c}$ the mean of the observed values:

$$APE_n = 100\frac{|c_n - \hat{c}_n|}{c_n}, MdAPE = \underset{n}{median}\, APE_n, MAPE = \frac{1}{N}\sum_{n=1}^{N} APE_n, \quad (S1)$$

$$MAE = \frac{1}{N}\sum_{n=1}^{N} |c_n - \hat{c}_n|, RMSE = \sqrt{\frac{1}{N}\sum_{n=1}^{N} (c_n - \hat{c}_n)^2}, \quad (S2)$$

$$R^2 = 1 - \frac{\sum_{n=1}^{N} (c_n - \hat{c}_n)^2}{\sum_{n=1}^{N} (c_n - \acute{c})^2}. \quad (S3)$$

APE is the absolute percentage error, MdAPE and MAPE its median and mean, MAE the mean absolute error, RMSE the root-mean-square error and R^2 the coefficient of determination. MAE and RMSE are in t CO_2 per person.

C. Recurrent, convolutional and attention layers

All deep networks receive the standardized window as a sequence $\tilde{x}_1, \ldots, \tilde{x}_W$ of covariate vectors, written x_τ below for brevity. In every equation, W and U with a subscript are weight matrices that multiply the current input and the previous hidden state, b with a subscript is a bias vector, $\sigma(\cdot)$ is the logistic function applied element by element, $\tanh(\cdot)$ is the hyperbolic tangent and $\odot$ is the element-by-element product. All weights and biases are learned in training.

Long short-term memory (LSTM). At window step τ the layer updates a cell state c_τ and a hidden state h_τ through a forget gate f_τ, an input gate i_τ, an output gate o_τ and a candidate cell state $\tilde{c}_\tau$:

$$f_\tau = \sigma(W_f x_\tau + U_f h_{\tau-1} + b_f), i_\tau = \sigma(W_i x_\tau + U_i h_{\tau-1} + b_i), \quad (S4)$$

$$o_\tau = \sigma\left(W_o x_\tau + U_o h_{\tau-1} + b_o\right), \tilde{c}_\tau = \tanh\left(W_c x_\tau + U_c h_{\tau-1} + b_c\right), \quad (S5)$$

$$c_\tau = f_\tau \odot c_{\tau-1} + i_\tau \odot \tilde{c}_\tau, h_\tau = o_\tau \odot \tanh\left(c_\tau\right). \quad (S6)$$

The forget gate decides how much of the previous cell state is kept, the input gate how much of the candidate is added, and the output gate how much of the cell state is passed on as the hidden state.

Gated recurrent unit (GRU). The GRU has no separate cell state. An update gate z_τ and a reset gate r_τ control the hidden state:

$$z_\tau = \sigma\left(W_z x_\tau + U_z h_{\tau-1} + b_z\right), r_\tau = \sigma\left(W_r x_\tau + U_r h_{\tau-1} + b_r\right), \quad (S7)$$

$$h_\tau = \left(1 - z_\tau\right) \odot h_{\tau-1} + z_\tau \odot \tanh\left(W_h x_\tau + U_h\left(r_\tau \odot h_{\tau-1}\right) + b_h\right). \quad (S8)$$

One-dimensional convolution. Convolutional layer *l* with kernel size *k* computes, at each window step, a weighted sum of the outputs of the previous layer at neighbouring steps and applies the rectified linear unit $ReLU(a) = max(0, a)$:

$$z_\tau^{(l)} = ReLU\left(\sum_{j=1}^{k} K_j^{(l)} z_{\tau+j-\lceil k/2 \rceil}^{(l-1)} + b^{(l)}\right), \quad (S9)$$

where $z_\tau^{(0)} = x_\tau$, $K_j^{(l)}$ is the j-th weight matrix of the kernel of layer *l*, $b^{(l)}$ its bias and $\lceil \cdot \rceil$ rounds up to the next integer.

Bidirectional LSTM. Two LSTM layers read the sequence forwards and backwards, and their hidden states $\vec{h}_\tau$ and $\acute{h}_\tau$ are joined into one vector:

$$h_\tau = \left[\vec{h}_\tau ; \acute{h}_\tau\right]. \quad (S10)$$

Additive attention pooling. The hidden states of the W steps are combined into one context vector *s* with weights α_τ that sum to one:

$$u_\tau = v^\top \tanh\left(W_a h_\tau + b_a\right), \alpha_\tau = \frac{\exp\left(u_\tau\right)}{\sum_{\tau'=1}^{W} \exp\left(u_{\tau'}\right)}, s = \sum_{\tau=1}^{W} \alpha_\tau h_\tau, \quad (S11)$$

where u_τ is the score of step τ, and the weight matrix W_a, the bias b_a and the vector v are learned. A weight α_τ of $1/W = 0.125$ for every step corresponds to a simple average.

Training loss. All networks minimize the mean squared error of y over the N_{tr} training samples:

$$L = \frac{1}{N_{tr}} \sum_{n=1}^{N_{tr}} \left(y_n - \hat{y}_n\right)^2. \quad (S12)$$

D. Particle swarm optimization

Six settings of KG-CBA were tuned (Table 2 of the article gives the search ranges and the selected values). Each of $J = 6$ particles has a position p_j, a vector of the six settings, and a velocity v_j. In each of $K = 5$ iterations, every particle is evaluated and then moved:

$$v_j \leftarrow \omega v_j + c_1 r_1 \odot \left(p_j^{best} - p_j\right) + c_2 r_2 \odot \left(g^{best} - p_j\right), p_j \leftarrow clip\left(p_j + v_j, l, u\right), \quad (S13)$$

where p_j^{best} is the best position found so far by particle j, g^{best} the best position found by any particle, $\omega = 0.7$ the inertia, $c_1 = c_2 = 1.5$ the cognitive and social constants, r_1 and r_2 vectors of independent random numbers between 0 and 1, l and u the lower and upper bounds of the settings, and $clip$ limits each setting to its bounds. Velocities are limited to 30% of the range of each setting.

Algorithm A. Particle swarm search for the settings of KG-CBA.

```
Input: bounds [l, u] for (filters, kernel size, LSTM units, dropout, log learning rate, batch
size);
       swarm size J = 6, iterations K = 5; country-disjoint training/validation split of fold 1
Initialize positions p_j ~ U(l, u) and velocities v_j ~ U(-1, 1) x 0.15 (u - l)
for k = 1..K:
    for j = 1..J:
        settings <- decode(p_j)              # round integers, exponentiate the learning rate
        build KG-CBA with these settings; start the Kaya head at the last-year ridge solution
        train with early stopping (patience 6, at most 60 epochs) on the inner validation
countries
        f_j <- smallest validation mean squared error
        update the personal best (p_j_best, f_j_best) and the global best (g_best, f_best)
```

```
    v_j <- 0.7 v_j + 1.5 r1 (p_j_best - p_j) + 1.5 r2 (g_best - p_j);  limit to 0.3 (u - l)
    p_j <- clip(p_j + v_j, l, u)
Output: g_best = (37 filters, kernel 4, 40 units, dropout 0.231, learning rate 9.96e-4, batch
48);
        f_best = 0.0522
```

E. Training of the Kaya-guided hybrid network

Algorithm B. Training of KG-CBA in one fold.

```
Input: training countries of fold k; inner split by country (6/7 for fitting, 1/7 for validation,
rounded down: 12 countries in Experiment A, 8 in Experiment C)
1. Standardize the windows with the training mean and standard deviation of each covariate
2. Fit ridge regression (lambda = 1) on the covariates of the most recent year -> (beta, beta_0)
3. Build the network:
     Kaya head: one linear unit on the most recent year, started at (beta, beta_0)
     residual branch: Conv1D x 2 -> bidirectional LSTM -> dropout -> attention pooling ->
Dense(32)
                      -> linear output started near zero (normal, standard deviation 0.001)
     estimate = Kaya head + residual branch
4. Train both parts together: Adam optimizer, mean squared error, at most 90 epochs,
   stop after 9 epochs without improvement on the inner validation countries
5. Estimate the held-out countries of fold k
```

F. Prediction intervals

Monte-Carlo dropout. With dropout active at prediction time, $T = 60$ predictions $\hat{y}^{(1)}, \ldots, \hat{y}^{(T)}$ of the same sample give the predictive mean and standard deviation

$$\dot{y} = \frac{1}{T}\sum_{s=1}^{T} \hat{y}^{(s)}, \hat{\sigma} = \sqrt{\frac{1}{T}\sum_{s=1}^{T} \left(\hat{y}^{(s)} - \dot{y}\right)^2}. \quad (S14)$$

Split-conformal calibration. For each of the n_{cal} samples of the inner validation countries of a fold, the score is

$$s = \frac{|y - \dot{y}|}{\hat{\sigma}}, \quad (S15)$$

and the calibration factor $\hat{q}$ is the $\lceil 0.9(n_{cal}+1) \rceil$-th smallest score. The 90% interval for a test sample, in t CO_2 per person, is

$$\left[\exp(\dot{y} - \hat{q}\,\hat{\sigma}), \exp(\dot{y} + \hat{q}\,\hat{\sigma})\right]. \quad (S16)$$

Calibration variants for ridge regression. Ridge regression gives no predictive standard deviation, so the score is the absolute error $s = |y - \hat{y}|$ and the 90% interval is

$[\exp(\hat{y}-\hat{q}), \exp(\hat{y}+\hat{q})]$. Three calibration sets were compared in each of the five folds. (a) *Calibration countries as for the hybrid.* One-seventh of the training countries, rounded down (12 in Experiment A, 8 in Experiment C), were drawn at random and set aside for calibration, and the model was fitted to the others; this was repeated for 200 random draws. (b) *All training countries.* An inner five-fold leave-country-out loop over the training countries (89 or 90 in Experiment A, 60 or 61 in Experiment C) gave an out-of-fold error for every training sample; these errors were the calibration scores, and the model fitted to all training countries produced the estimates. (c) *Exchangeable reference.* The out-of-fold errors of the held-out countries themselves were divided at random into a calibration half and an evaluation half (200 repetitions); calibration and evaluation samples then come from the same countries, so that the exchangeability assumption holds. Results are given in S2 Table.

G. SHAP values

For a fitted model f and one sample, the SHAP value of input j is

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!} \left[f(S \cup \{j\}) - f(S) \right], \quad (S17)$$

where F is the set of all inputs (here the $W \times F = 56$ values of the window in Experiment C), S is a subset of inputs that does not contain j, $|S|$ is the number of inputs in S, $|F|$ the total number of inputs, and $f(S)$ the expected estimate of the model when only the inputs in S are known. The fraction is the weight of subset S, and the term in square brackets is the change in the estimate when input j is added to S. The SHAP values of a sample sum to the difference between its estimate and the mean estimate. A positive ϕ_j means that input j raises the estimate of y for that sample, a negative value that it lowers it, and the mean of $|\phi_j|$ over samples measures the overall

importance of the input. Importance by covariate aggregates the values of the eight years of that covariate, and importance by year aggregates the values of the seven covariates of that year.

H. Country-clustered comparison of two models

Algorithm C. Paired comparison with countries as clusters.

```
Input: held-out absolute percentage errors a and b of two models for the same N country-years;
       country label g of every country-year
1. p_obs     <- Wilcoxon signed-rank test on (a, b)            # observation level; for contrast
only
2. A_c <- median of a in country c;  B_c <- median of b in country c
   p_country <- Wilcoxon signed-rank test on (A, B) over the countries
3. delta     <- median of (a - b) over all country-years
   for r = 1..10000:
       draw as many countries as there are, with replacement, keeping all years of a drawn
country
       delta_r <- median of (a - b) over the drawn country-years
   CI     <- 2.5th and 97.5th percentiles of delta_r
   p_boot <- 2 x min( share of delta_r <= 0, share of delta_r >= 0 )
Output: p_obs, p_country, delta, CI, p_boot; conclusions follow p_boot
```

I. Folds and per-fold results

The five folds contain 23, 23, 22, 22 and 22 countries; the fold of every country is given in S1 Table. Table B gives the median absolute percentage error of the leading models in each fold. The differences between folds are larger than the differences between models, which is why models are compared by paired tests with countries as clusters and not by fold averages.

Table B. Median absolute percentage error (%) by fold, Experiment A.

Fold	Countries	Ridge regression	GRU	PSO-KG-CBA	Ensemble
1	23	11.6	15.7	11.7	10.1
2	23	23.2	29.5	24.9	25.6
3	22	18.1	18.4	17.7	18.3
4	22	15.8	14.2	17.6	13.7
5	22	13.8	17.3	13.0	10.7